

Name: M. Akhil kumar

Reg no: 192572060

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Course name: Computer Vision

Scenario 1: Manufacturing Quality Inspection

(a) Interpret the scenario and identify the type of noise present.

The image contains periodic noise caused by electrical interference, appearing as repetitive horizontal stripes.

(b) Explain how this degradation affects inspection accuracy.

Periodic noise hides defects, reduces image clarity, and may lead to incorrect identification of cracks or surface flaws during quality inspection.

(c) Recommend an appropriate enhancement technique to remove the interference.

Use the Fourier Transform (Frequency Domain) with a Notch filter to remove the periodic noise.

(d) Justify why your selected approach is more suitable than spatial-domain methods.

Periodic noise appears at specific frequencies in the Fourier domain. A notch filter removes

only those unwanted frequencies while preserving important image details. Spatial filters cannot selectively remove periodic noise and may blur the entire image.

(e) Explain your answer in a logical sequence.

- * Capture the noisy image.
- * Apply the Fourier Transform.
- * Detect the frequency peaks caused by periodic noise.
- * Apply a notch filter to remove those frequencies.
- * Perform the inverse Fourier Transform to obtain a clean image for accurate defect detection.

Scenario 2: Autonomous Vehicle Navigation

(a) Interpret the scenario and explain the challenge involved.

The vehicle captures images in heavy rain, introducing noise while lane markings and road edges must remain clear for safe

navigation.

(b) Identify the importance of preserving edges while removing noise.

Edges represent lanes, road boundaries, pedestrians, and obstacles. Preserving them ensures accurate detection and safe vehicle control.

(c) Recommend a suitable Filtering technique. Use a Bilateral Filter.

(d) Justify why your selected Filter preserves important image details.

A bilateral filter smooths noisy regions while preserving edges by considering both pixel distance and intensity differences. Unlike Gaussian Filtering, it reduces noise without blurring important boundaries.

(e) Present your explanation clearly and systematically.

*Capture the noisy road image.

*Apply bilateral Filtering.

left;">*Remove rain noise while preserving lane edges.

*Detect lane markings and road boundaries accurately.

*Use the enhanced image for safe navigation and decision-making.